DOI: 10.1002/adsc.201200910

Optically Active 2,3-Allenoates *via* Palladium-Catalyzed Carbonylation of Enantioenriched Propargylic Mesylates

Yuli Wang[a] and Shengming Ma[a,b,*]

[a] Shanghai Key Laboratory of Green Chemistry and Chemical Process, Department of Chemistry, East China Normal University, 3663 North Zhongshan Lu, Shanghai 200062, People's Republic of China
Fax: (+86) 21-62609305; e-mail: masm@sioc.ac.cn
[b] State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, 345 Lingling Lu, Shanghai 200032, People's Republic of China

Received: October 11, 2012; Revised: December 1, 2012; Published online: February 22, 2013

Supporting information for this article is available on the WWW under http://dx.doi.org/10.1002/adsc.201200910.

Abstract: Under mild conditions, optically active 2,3-allenoates were synthesized from enantioenriched propargylic mesylates with moderate to excellent yields and high efficiency of chirality transfer by using palladium(0) bis(dibenzylideneacetone) (3 mol%) with (*S*)-(−)-5,5′-bis(diphenylphosphino)-4,4′-bi-1,3-benzodioxole [(*S*)-SEGPHOS] (3 mol%) as the catalyst and diammonium hydrogen phosphate (1.1 equiv.) as the base.

Keywords: carbonylation; chirality transfer; optically active allenoates; palladium; propargylic mesylates

Introduction

2,3-Allenoates are a class of interesting and versatile synthetic intermediates, which may undergo many reactions,[1] such as nucleophilic addition reactions,[2] electrophilic addition reactions,[1f,3] Morita–Baylis–Hillman reactions,[4] rearrangements,[5] cycloaddition[6] and cyclization reactions,[1c,7] etc.[8] Many methods have been reported for the synthesis of allenoates,[9] but the preparation of optically active 2,3-allenoates is not very well established (Scheme 1).[10–18] Use of stoichiometric chiral reagents and/or limited substrate scopes are their main drawbacks.

Central-to-axial chirality transfer has been proven to be an efficient strategy for the synthesis of optically active allenes. In this field, Marshall et al. reported the synthesis of chiral allenoates by using enantioenriched propargylic mesylates, which is very practical for the *terminal* propargylic mesylates. However, serious racemization was observed for internal propargylic mesylates (Scheme 2).[17]

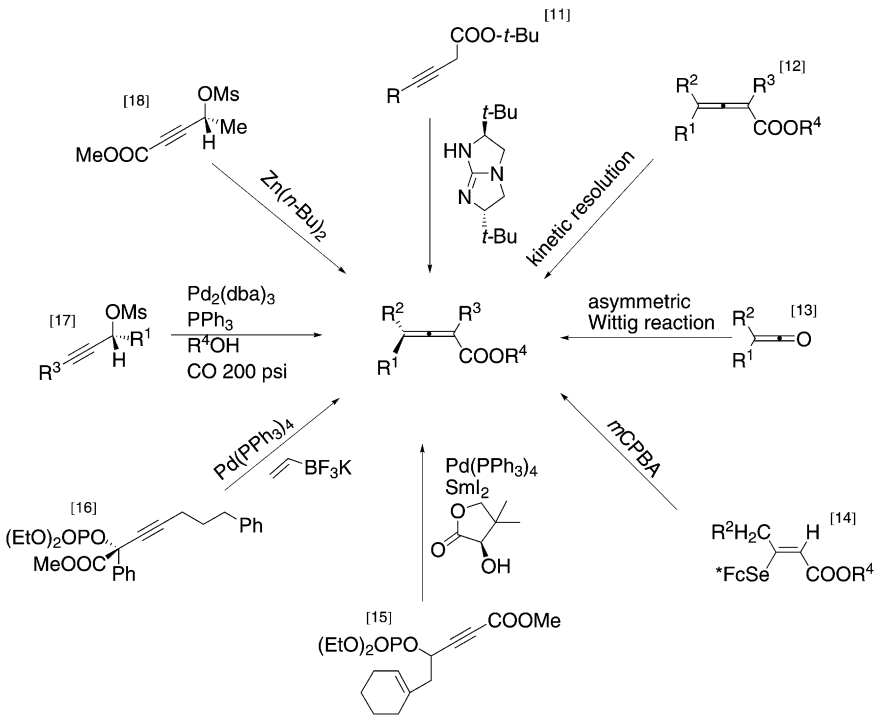
Herein, we report a palladium-catalyzed carbonylation of enantioenriched *non-terminal* propargylic mesylates for the preparation of optically active 2,4-disubstituted 2,3-allenoates using $Pd(dba)_2$ and (*S*)-SEGPHOS as the catalyst.

Results and Discussion

We tried the reaction of enantioenriched (*S*)-1-ethyl-2-heptynyl methanesulfonate [(*S*)-**1a**]. Unfortunately, the efficiency of chirality transfer was rather low when PPh_3 was used as the ligand (Table 1, entry 1). Then several optically active phosphorus ligands were examined with the purpose to prevent racemization. (*S*)-SEGPHOS provided the best results: the optical purity of the product dropped to 96% from the enantioenriched starting material with 99% *ee*, and the yield was 70% (Table 1, entries 2–5). When (*R*)-SEGPHOS was used, the efficiency of chirality transfer dropped slightly, however, the reaction was very slow (Table 1, entry 6). It is obvious that (*S*)-SEGPHOS matches the substrate (*S*)-**1a**, while the *R* isomer does not. $Pd(acac)_2$ also works with a lower yield (Table 1, entry 7). Meanwhile, $[PdCl(\pi\text{-allyl})]_2$ is much less effective (Table 1, entry 8).

Solvent screening indicated that an ether is the type of better solvent: Methyl *tert*-butyl ether gave the best result (see the Supporting Information for details on solvent effects). Several bases were also tested in order to improve the yield and efficiency of chirality transfer. Most of the bases, such as TEA, pyridine, DIPEA and $NaHCO_3$ failed to improve the yields with different stereoselectivity (Table 2, entries 1–4); interestingly, when $(NH_4)_2HPO_4$ was used, the yield was improved obviously, and the *ee* value of

 © 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

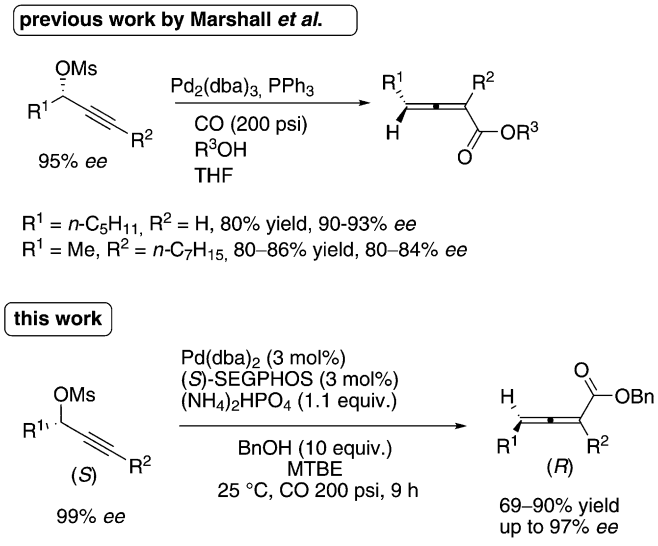


Scheme 1. Method for synthesis of chiral 2,3-allenoates.

the product was kept at the level of 95% (Table 2, entry 5).

Under our standard conditions, propargylic carbonate could not form 2,3-allenoate; propargylic phosphate and tosylate could lead to the desired product in moderate yield, but the enantiopurity of the product was poor. The reaction occurs smoothly with many alcohols, and all the different allenoates were obtained in good yields with very different *ee*s: When methanol was used, the *ee* value of the product was 80%; the enantiopurity dropped sharply in the presence of ethanol (84%); isobutyl alcohol performed better with 90% *ee* (Scheme 3). In conclusion, 3 mol% $Pd(dba)_2$, 3 mol% (*S*)-SEGPHOS, 1.1 equiv. of $(NH_4)_2HPO_4$, 10 equiv. of BnOH and 2 mL of MTBE were defined as our standard conditions.

Next, the optimized reaction conditions were applied to a variety of substrates (Table 3). Firstly, we fixed R^1 to ethyl to examine the effect of the R^2 group. When R^2 is the smaller methyl group, the enantiopurity dropped to 90% (Table 3, entry 2); when R^2 became larger, such as ethyl, propyl, hexyl, *tert*-butyl and isopentyl, 2,3-allenoates could be prepared in good yields with higher *ee* (94%~97%) (Table 3, entries 3–7), although the substrate with an isopentyl group a required longer reaction time; introducing a chlorine atom into the aliphatic alkyl group has a very limited impact on the reaction (Table 3, entry 8); when unsaturated groups, such as phenyl and allyl groups, were introduced into R^2, the efficiency of chirality transfer dropped slightly (Table 3, entries 9 and 10); when R^2 is the 2-methylallyl group, the corresponding 2,3-allenoate (R_a)-**2k** was obtained with 95% *ee* (Table 3, entry 11). Different R^1 groups were also screened. With a longer chain, the enantiopurity of the product dropped slightly (Table 3, entry 12). The more sterically hindered substrates (*S*)-**1m** and (*S*)-**1n** required longer reaction times, but chirality transfer proceeded smoothly (Table 3, entries 13 and 14). An allyl group can also be introduced to R^1 with no influence on the

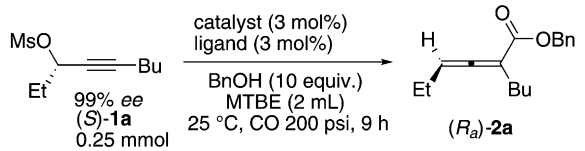


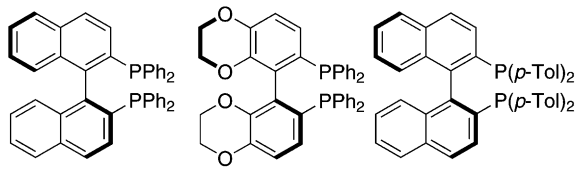
Scheme 2. Comparison of previous works and this work.

 © 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

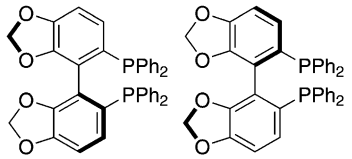
16154169, 2013, 4, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201200910 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 1. The effect of catalyst and ligand on Pd-catalyzed carbonylation of enantioenriched propargylic mesylate (*S*)-**1a**.[a]





(*S*)-BINAP (*S*)-SYNPHOS (*S*)-**L**



(*S*)-SEGPHOS (*R*)-SEGPHOS

Entry	Catalyst	Ligand	Yield [%][b] (*ee* [%][c]) of (*R*$_a$)-**2a**
1	$Pd(dba)_2$	PPh_3[d]	78 (66)
2	$Pd(dba)_2$	(*S*)-BINAP	70 (90)
3	$Pd(dba)_2$	(*S*)-SYNPHOS	74 (88)
4	$Pd(dba)_2$	(*S*)-**L**	63 (93)
5	$Pd(dba)_2$	(*S*)-SEGPHOS	70 (96)
6	$Pd(dba)_2$	(*R*)-SEGPHOS	35[e] (93)
7	$Pd(acac)_2$	(*S*)-SEGPHOS	54 (96)
8	[PdCl (π-allyl)]$_2$	(*S*)-SEGPHOS	32 (22)

[a] MTBE = methyl *tert*-butyl ether.
[b] Isolated yield.
[c] The *ee* value of the product was determined by HPLC.
[d] 6 mol% of PPh_3 was used.
[e] 56% of the starting material was remained, determined by ^{1}H NMR analysis using 1,3,5-trimethylbenzene as the internal standard.

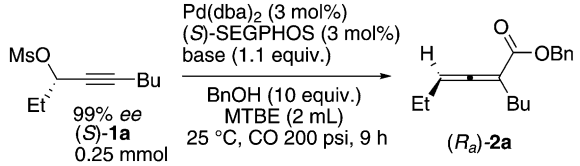
reaction (Table 3, entry 15). As a comparison, the same substrate (*S*)-**1p** used by Marshall et al.[17] performs better under the current conditions: the corresponding 2,3-allenoate (*R*$_a$)-**2p** was isolated in 93% *ee* from (*S*)-1-methyl-2-decynyl methanesulfonate [(*S*)-**1p**] with 99% *ee* (Table 3, entry 16).

2,3-Allenoate (*R*$_a$)-**2b** could be converted to the corresponding iodobutenolide (*R*)-**3** *via* iodolactonization with I_2 (Scheme 4).[19]

The absolute configuration of (*R*)-**3** was determined by an X-ray diffraction study.[20] Based on this result, we assigned the configuration of the 2,3-allenoate prepared from (*S*)-**1b** as *R* (Figure 1).

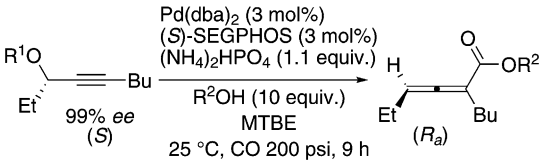
A working model to predict the absolute configuration of the 2,3-allenoate tentatively assigned for this

Table 2. Selected results with different bases.[a]



Entry	Base	Yield [%] of (*R*$_a$)-**2a**[b]	*ee* [%] of (*R*$_a$)-**2a**[c]
1	TEA	70	91
2	pyridine	72	84
3	DIPEA	71	91
4	$NaHCO_3$	64	91
5	$(NH_4)_2HPO_4$	81	95

[a] TEA = triethylamine, DIPEA = ethyldiisopropylamine.
[b] Isolated yield.
[c] The *ee* value of the product was determined by HPLC.

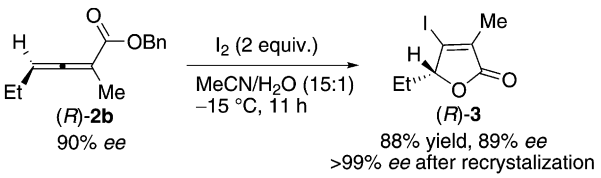


R^1 = COOMe, R^2 = Bn, yield = 2%[a]
R^1 = P(O)(OEt)$_2$, R^2 = Bn, yield = 74%, *ee* = 57%
R^1 = Ts , R^2 = Bn, yield = 71%, *ee* = 66%
R^1 = Ms, R^2 = Me, yield = 71%, *ee*[b] = 80%
R^1 = Ms, R^2 = Et, yield = 73%, *ee*[b] = 84%
R^1 = Ms, R^2 = *i*-Bu yield = 79%, *ee*[b] = 90%

[a] 89% of the starting material was recovered as determined by ^{1}H NMR analysis using 1,3,5-trimethylbenzene as the internal standard.
[b] The *ee* values were determined by the corresponding iodobutenolides which accomplished *via* iodolactonization of the products with I_2.

Scheme 3. Results with different leaving groups and alcohols.

reaction is shown in Scheme 5.[21] The allenylpalladium species **A** is formed after oxidative addition of the enantioenriched propargylic mesylate to Pd(0) species. Subsequently CO is activated by the coordination with palladium. Nucleophilic attack of the alcoholic oxygen on the coordinated CO in the presence of base leads to the alkoxycarbonyl intermediate **C**. Reductive elimination would afford the final product (path a). Since organopalladium compounds are known to undergo a carbon monoxide insertion reac-



Scheme 4. Iodolactonization of the 2,3-allenoate (*R*$_a$)-**2b**.

 © 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2013, 4, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201200910 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

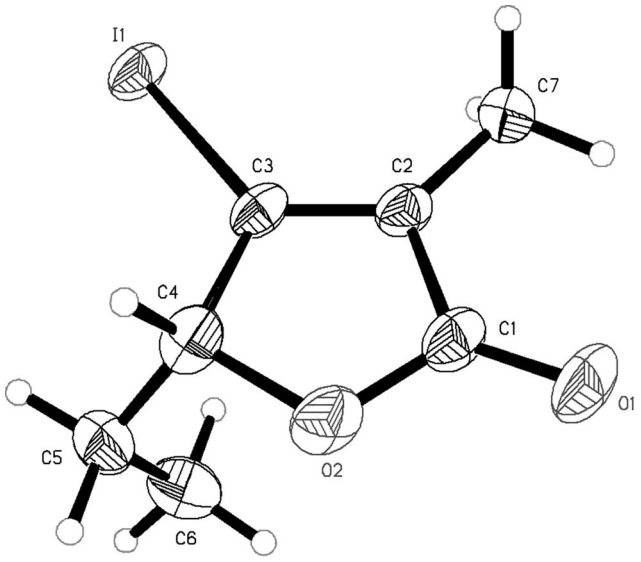
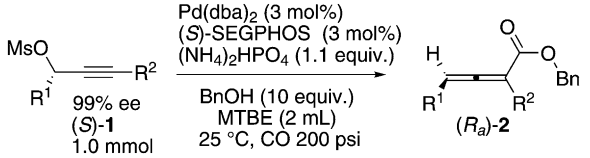


Figure 1. ORTEP representation of (*R*)-**3**.

tion very readily, a CO insertion to the intermediate **A** followed by attack of the alcohol in the presence of base would lead to the final product *via* acylpalladium intermediate **E** (path b). In the presence of (*S*)-SEGPHOS, the chirality in substrates with the *S* configuration should be transferred to the products with R_a configuration effectively *via* the intermediacy of optically active intermediate **C**. Partial racemization may be caused by the isomerization between the allenyl palladium species and propargylic palladium intermediates.

Table 3. Substrate scope for Pd-catalyzed carbonylation of enantioenriched propargylic mesylate.



Entry	(*S*)-**1:** R^1, R^2	Time [h]	Yield [%][a] (*ee* [%][b]) of (R_a)-**2**
1	Et, Bu (**1a**)	9	81 (**2a**) (96)
2	Et, Me (**1b**)	9	72 (**2b**) (90)
3	Et, Et (**1c**)	9	86 (**2c**) (97)
4	Et, Pr (**1d**)	9	84 (**2d**) (97)
5	Et, C_6H_{13} (**1e**)	9	80 (**2e**) (95)
6	Et, *t*-Bu (**1f**)	9	79 (**2f**) (97)
7	Et, *i*-Pent (**1g**)	12	90 (**2g**) (94)
8	Et, $(CH_2)_4Cl$ (**1h**)	9	88 (**2h**) (95)
9	Et, $(CH_2)_2Ph$ (**1i**)	9	84 (**2i**) (92)
10	Et, allyl (**1j**)	9	85 (**2j**) (91)
11	Et, 2-Meallyl (**1k**)	9	88 (**2k**) (95)
12	C_7H_{15}, Bu (**1l**)	9	85 (**2l**) (94)
13	*i*-Bu, Bu (**1m**)	10.5	86 (**2m**) (97)
14	Bn, Bu (**1n**)	12	69 (**2n**) (96)
15	allyl, Bu (**1o**)	9	81 (**2o**) (96)
16	Me, C_7H_{15} (**1p**)	9	85 (**2p**) (93)

[a] Isolated yield.
[b] The *ee* value of the product was determined by HPLC.

Conclusions

In conclusion, we have developed an efficient method for the synthesis of optically active 2,3-allenoates from the readily available enantioenriched secondary propargylic mesylates with moderate to excellent yields and good stereoselectivities under very mild reaction conditions by using the palladium catalysis and (*S*)-SEGPHOS as the ligand: Various alkyl, allyl, or benzyl substituents may be successfully introduced into the substrates. Further research in this area is being pursued in our laboratory.

Experimental Section

General Information

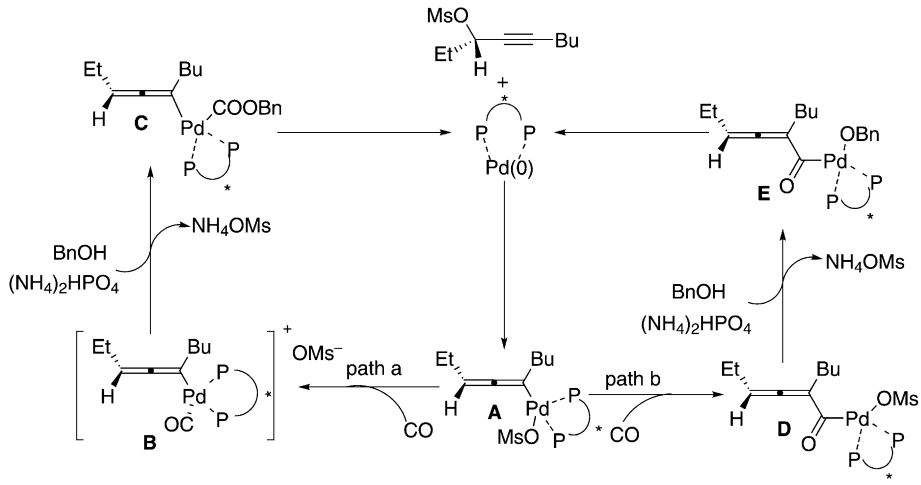
NMR spectra were taken with a Bruker-300 spectrometer (300 MHz for ^{1}H NMR, 75.4 MHz for ^{13}C NMR) or Bruker-400 spectrometer (400 MHz for ^{1}H NMR, 100.5 MHz for ^{13}C NMR) in $CDCl_3$. Chemical shifts were recorded in ppm relative to the residue of $CHCl_3$ in $CDCl_3$ and coupling constants were reported in Hz. X-ray intensity data were measured at 293 K on a Bruker SMART APEX CCD-based X-ray diffractometer system equipped with a fine-focus Mo-target X-ray tube ($\lambda = 0.71073$ Å) operated at 1,500 W. All reactions were carried out in oven-dried tubes. $(NH_4)_2HPO_4$was purchased from Sinopharm Chemical Reagent Co., Ltd; (*S*)-SEGPHOS was purchased from Stream Chemicals. MTBE was dried over sodium wire with benzophenone as indicator and distilled freshly before use. The enantioenriched propargylic mesylates were synthesized from the corresponding enantiopure propargylic alcohols, and used without further purification. Purification by column chromatography was performed using Haiyang (Shandong, China) silica gel (10–40 μL).

Preparation of (R_a)-Benzyl 2-Butyl-2,3-hexadienoate [(R_a)-2a]; Typical Procedure

To a flame-dried tube were added $Pd(dba)_2$ (17.3 mg, 0.03 mmol), (*S*)-SEGPHOS (18.5 mg, 0.03 mmol), $(NH_4)_2HPO_4$ (144.7 mg, 1.1 mmol), (*S*)-**1a** (218.6 mg, 1.0 mmol)/MTBE (1 mL) and BnOH (1.0798 g, 10 mmol)/MTBE (1 mL). The tube was then placed into the Parr reactor. After the air in the reactor had been replaced by CO gas three times sequentially, the Parr reactor was charged to 200 psi with CO gas. The mixture was stirred for 9 h at room temperature, then the gas was vented in the hood safely. The solution was diluted with Et_2O, washed with brine (10 mL × 3), and dried with anhydrous Na_2SO_4. After filtra-

 © 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2013, 4, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201200910 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License



Scheme 5. Proposed mechanism and prediction of the absolute configuration of the allenoate.

tion and concentration under reduced pressure, the crude product was purified by flash chromatography on silica gel (eluent: petroleum ether/ethyl ether = 30/1) to afford (R_a)-**2a** as a liquid; yield: 210.4 mg (81%); 96% *ee*; HPLC [conditions: Regis (*S*,*S*) Whelk-O column, hexane/*i*-PrOH = 200/1, 0.5 mL min^{-1}]: λ = 214 nm, t_R(minor) = 19.4 min, t_R-(major) = 22.2 min; $[\alpha]_D^{24}$: −38.0 (*c* 1.17, $CHCl_3$); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.40–7.26 (m, 5H, Ar-H), 5.63–5.55 (m, 1H, =CH), 5.22 (d, *J* = 12.6 Hz, 1H, one proton from Bn), 5.15 (d, *J* = 12.9 Hz, 1H, one proton from Bn), 2.28–2.20 (m, 2H, CH_2), 2.19–2.07 (m, 2H, CH_2), 1.49–1.28 (m, 4H, 2CH_2), 1.05 (t, *J* = 7.4 Hz, 3H, CH_3), 0.90 (t, *J* = 7.1 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 209.7, 167.5, 136.4, 128.3, 127.7, 127.4, 101.1, 96.6, 66.0, 30.2, 28.1, 22.1, 21.3, 13.8, 13.1; MS (ESI): *m/z* = 281 $(M+Na)^+$, 259 $(M+H)^+$; IR (neat): ν = 3091, 3066, 3034, 2960, 2931, 2872, 1955, 1709, 1455, 1376, 1263, 1236, 1215, 1119, 1058, 1012 cm^{-1}; HR-MS: *m/z* = 258.1624, calcd. for $C_{17}H_{22}O_2$ $[M^+]$: 258.1620.

The following compounds [(R_a)-**2b**-(R_a)–**2p**] in Table 3 as well as (R_a)-methyl 2-butyl-2,3-hexadienoate, (R_a)-ethyl 2-butyl-2,3-hexadienoate and (R_a)-isobutyl 2-butyl-2,3-hexadienoate in Scheme 3) were prepared according to the above Typical Procedure.

(R_a)-Benzyl 2-methyl-2,3-hexadienoate [(R_a)-2b]: The reaction of $Pd(dba)_2$ (17.3 mg, 0.03 mmol), (*S*)-SEGPHOS (18.6 mg, 0.03 mmol), $(NH_4)_2HPO_4$ (145.1 mg, 1.1 mmol), (*S*)-**1b** (176.3 mg, 1.0 mmol)/MTBE (1 mL), BnOH (1.0806 g, 10 mmol)/MTBE (1 mL) afforded (R_a)-**2b** (eluent: petroleum ether/ethyl ether = 30/1) as a liquid; yield: 155.5 mg (72%); 90% *ee;* HPLC [conditions: Regis (*S*,*S*) Whelk-O column, hexane/*i*-PrOH = 200/1, 0.5 mL min^{-1}, λ = 214 nm]: t_R (minor) = 22.4 min, t_R (major) = 25.5 min; $[\alpha]_D^{20}$: −65.6 (*c* 1.17, $CHCl_3$); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.40–7.26 (m, 5H, Ar-H), 5.58–5.49 (m, 1H, =CH), 5.22 (d, *J* = 12.6 Hz, 1H, one proton from Bn), 5.15 (d, *J* = 12.9 Hz, 1H, one proton from Bn), 2.18–2.06 (m, 2H, CH_2 from Et), 1.89 (d, *J* = 3.0 Hz, 3H, CH_3 from Me), 1.04 (t, 3H, *J* = 7.5 Hz, CH_3 from Et); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 210.1, 167.9, 136.4, 128.4, 127.9, 127.6, 96.1, 95.7, 66.2, 21.3, 15.3, 13.2; MS (ESI): *m/z* = 239 $(M+Na)^+$; IR (neat): ν = 3090, 3066, 3034, 2966, 2932, 2875, 1958, 1710, 1455, 1375, 1266, 1114 cm^{-1}; HR-MS: *m/z* = 216.1151, calcd. for $C_{14}H_{16}O_2$ $[M^+]$: 216.1150.

(R_a)-Benzyl 2-ethyl-2,3-hexadienoate [(R_a)-2c]: The reaction of $Pd(dba)_2$ (17.3 mg, 0.03 mmol), (*S*)-SEGPHOS (18.5 mg, 0.03 mmol), $(NH_4)_2HPO_4$ (145.1 mg, 1.1 mmol), (*S*)-**1c** (189.7 mg, 1.0 mmol)/MTBE (1 mL), BnOH (1.0801 g, 10 mmol)/MTBE (1 mL) afforded (R_a)-**2c** (eluent: petroleum ether/ethyl ether = 30/1) as a liquid; yield: 197.3 mg (86%); 97% *ee*; HPLC [conditions: Regis (*S*,*S*) Whelk-O column, hexane/*i*-PrOH = 200/1, 0.5 mL min^{-1}, λ = 214 nm]: t_R (minor) = 20.1 min, t_R (major) = 23.2 min; $[\alpha]_D^{26}$: −55.9 (*c* 1.22, $CHCl_3$); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.41–7.27 (m, 5H, Ar-H), 5.69–5.59 (m, 1H, =CH), 5.23 (d, *J* = 12.6 Hz, 1H, one proton from Bn), 5.16 (d, *J* = 12.6 Hz, 1H, one proton from Bn), 2.38–2.20 (m, 2H, CH_2), 2.20–2.07 (m, 2H, CH_2), 1.05 (t, *J* = 7.4 Hz, 3H, CH_3), 1.04 (t, *J* = 7.4 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 209.5, 167.5, 136.4, 128.4, 127.8, 127.5, 102.9, 97.4, 66.1, 21.6, 21.3, 13.2, 12.5; MS (ESI): *m/z* = 483 $(2M+Na)^+$, 253 $(M+Na)^+$, 231 $(M+H)^+$; IR (neat): ν = 3091, 3066, 3034, 2967, 2934, 2875, 1956, 1709, 1609, 1587, 1456, 1243, 1127, 1099, 1049, 1011 cm^{-1}; HR-MS: *m/z* = 230.1305, calcd. for $C_{15}H_{18}O_2$ $[M^+]$: 230.1307.

(R_a)-Benzyl 2-propyl-2,3-hexadienoate [(R_a)-2d]: The reaction of $Pd(dba)_2$ (17.3 mg, 0.03 mmol), (*S*)-SEGPHOS (18.5 mg, 0.03 mmol), $(NH_4)_2HPO_4$ (145.0 mg, 1.1 mmol), (*S*)-**1d** (203.4 mg, 1.0 mmol)/MTBE (1 mL), BnOH (1.0802 g, 10 mmol)/MTBE (1 mL) afforded (R_a)-**2d** (eluent: petroleum ether/ethyl ether = 30/1) as a liquid; yield: 205.2 mg (84%); 97% *ee*; HPLC [conditions: Regis (*S*,*S*) Whelk-O column, hexane/*i*-PrOH = 200/1, 0.5 mL min^{-1}, λ = 214 nm]: t_R (minor) = 18.2 min, t_R (major) = 20.8 min); $[\alpha]_D^{22}$: −44.2 (*c* 1.08, $CHCl_3$); ^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.41–7.26 (m, 5H, Ar-H), 5.63–5.56 (m, 1H, =CH), 5.21 (d, *J* = 12.8 Hz, 1H, one proton from Bn), 5.16 (d, *J* = 12.8 Hz, 1H, one proton from Bn), 2.30–2.18 (m, 2H, CH_2), 2.18–2.08 (m, 2H, CH_2), 1.53–1.41 (m, 2H, CH_2), 1.05 (t, *J* = 7.4 Hz, 3H, CH_3), 0.94 (t, *J* = 7.2 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 209.9, 167.6, 136.4, 128.4, 127.8, 127.5, 100.9, 96.7, 66.1, 30.5, 21.4, 13.6, 13.3; MS (EI, 70 eV): *m/z* (%) = 244 (M^+, 1.16), 91 (100); IR (neat): ν = 3066, 3034, 2963, 2933, 2873, 1956, 1709, 1498, 1456, 1376, 1230, 1191,

 © 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2013, 4, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201200910 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

1126, 1059, 1029 cm^{-1}; HR-MS: $m/z = 244.1464$, calcd. for $C_{16}H_{20}O_2$ [M^+]: 244.1463.

(R_a)-Benzyl 2-(*n*-hexyl)-2,3-hexadienoate [(R_a)-2e]: The reaction of $Pd(dba)_2$ (17.3 mg, 0.03 mmol), (*S*)-SEGPHOS (18.4 mg, 0.03 mmol), $(NH_4)_2HPO_4$ (145.1 mg, 1.1 mmol), (*S*)-**1e** (246.2 mg, 1.0 mmol)/MTBE (1 mL), BnOH (1.0807 g, 10 mmol)/MTBE (1 mL) afforded (R_a)-**2e** (eluent: petroleum ether/ethyl ether = 30/1) as a liquid; yield: 228.8 mg (80%); 95% *ee*; HPLC [conditions: Regis (*S*,*S*) Whelk-O column, hexane/*i*-PrOH = 200/1, 0.5 mL min^{-1}, λ = 214 nm]: t_R (minor) = 18.6 min, t_R (major) = 21.4 min; $[\alpha]_D^{24}$: −33.3 (*c* 1.21, $CHCl_3$); ^1H NMR (300 MHz, $CDCl_3$): δ = 7.40–7.27 (m, 5H, Ar-H), 5.63–5.54 (m, 1H, =CH), 5.22 (d, *J* = 12.9 Hz, 1H, one proton from Bn), 5.15 (d, *J* = 12.9 Hz, 1H, one proton from Bn), 2.30–2.18 (m, 2H, CH_2), 2.18–2.06 (m, 2H, CH_2), 1.50–1.20 (m, 8H, 4CH_2), 1.04 (t, *J* = 7.5 Hz, 3H, CH_3), 0.88 (t, *J* = 6.8 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 209.7, 167.5, 136.4, 128.3, 127.7, 127.4, 101.1, 96.6, 66.0, 31.6, 28.7, 28.4, 28.0, 22.5, 21.3, 14.0, 13.2; MS (ESI): m/z = 309 ($M+Na)^+$; IR (neat): ν = 3067, 3034, 2959, 2929, 2858, 1956, 1711, 1498, 1456, 1408, 1376, 1245, 1222, 1191, 1125, 1080, 1029, 1013 cm^{-1}; HR-MS: m/z = 286.1934, calcd. for $C_{19}H_{26}O_2$ [M^+]: 286.1933.

(R_a)-Benzyl 2-(*tert*-butyl)-2,3-hexadienoate [(R_a)-2f]: The reaction of $Pd(dba)_2$ (17.2 mg, 0.03 mmol), (*S*)-SEGPHOS (18.4 mg, 0.03 mmol), $(NH_4)_2HPO_4$ (144.9 mg, 1.1 mmol), (*S*)-**1f** (218.2 mg, 1.0 mmol)/MTBE (1 mL), BnOH (1.0790 g, 10 mmol)/MTBE (1 mL) afforded (R_a)-**2f** (eluent: petroleum ether/ethyl ether = 30/1) as a liquid; yield: 203.7 mg (79%); 97% *ee*; HPLC [conditions: Regis (*S*,*S*) Whelk-O column, hexane/*i*-PrOH = 99/1, 0.7 mL min^{-1}, λ = 214 nm]: t_R (minor) = 17.4 min, t_R (major) = 19.0 min; $[\alpha]_D^{23}$: −38.4 (*c* 1.01, $CHCl_3$); ^1H NMR (400 MHz, $CDCl_3$): δ = 7.39–7.27 (m, 5H, Ar-H), 5.56 (t, *J* = 6.2 Hz, 1H, =CH), 5.19 (d, *J* = 12.8 Hz, 1H, one proton from Bn), 5.13 (d, *J* = 12.8 Hz, 1H, one proton from Bn), 2.16–2.07 (m, 2H, CH_2 from Et), 1.20 (s, 9H, 3 × CH_3 from *t*-Bu), 1.03 (t, *J* = 7.2 Hz, 3H, CH_3 from Et); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 208.7, 166.6, 136.6, 128.4, 127.7, 127.4, 110.1, 96.6, 65.6, 33.0, 29.5, 21.3, 13.0; MS (ESI): m/z = 281 ($M+Na)^+$, 259 ($M+H)^+$; IR (neat): ν = 3091, 3067, 3034, 2963, 2934, 2907, 2870, 1950, 1709, 1457, 1362, 1256, 1217, 1050, 1025 cm^{-1}; HR-MS: m/z = 258.1614, calcd. for $C_{17}H_{22}O_2$ [M^+]: 258.1620.

(R_a)-Benzyl 2-(3-methylbutyl)-2,3-hexadienoate [(R_a)-2g]: The reaction of $Pd(dba)_2$ (17.3 mg, 0.03 mmol), (*S*)-SEGPHOS (18.6 mg, 0.03 mmol), $(NH_4)_2HPO_4$ (145.2 mg, 1.1 mmol), (*S*)-**1g** (232.2 mg, 1.0 mmol)/MTBE (1 mL), BnOH (1.0801 g, 10 mmol)/MTBE (1 mL) afforded (R_a)-**2g** (eluent: petroleum ether/ethyl ether = 30/1) as a liquid; yield: 244.8 mg (90%); 94% *ee*; HPLC [conditions: Regis (*S*,*S*) Whelk-O column, hexane/*i*-PrOH = 200/1, 0.5 mL min^{-1}, λ = 214 nm]: t_R (minor) = 17.5 min, t_R (major) = 19.8 min; $[\alpha]_D^{22}$: −33.0 (*c* 1.01, $CHCl_3$); ^1H NMR (300 MHz, $CDCl_3$): δ = 7.40–7.26 (m, 5H, Ar-H), 5.64–5.55 (m, 1H, =CH), 5.23 (d, *J* = 12.6 Hz, 1H, one proton from Bn), 5.16 (d, *J* = 12.9 Hz, 1H, one proton from Bn), 2.32–2.19 (m, 2H, CH_2), 2.19–2.07 (m, 2H, CH_2), 1.68–1.52 (m, 1H, CH), 1.39–1.26 (m, 2H, CH_2), 1.05 (t, *J* = 7.5 Hz, 3H, CH_3), 0.90 (d, *J* = 6.6 Hz, 6H, 2 × CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 209.8, 167.6, 136.5, 128.4, 127.8, 127.5, 101.4, 96.8, 66.1, 37.2, 27.6, 26.4, 22.50, 22.48, 21.4, 13.3; MS (ESI): m/z = 567 ($2M+Na)^+$, 295 ($M+Na)^+$, 273 ($M+H)^+$; IR (neat): ν = 3066, 3034, 2957, 2932, 2870, 1956, 1710, 1498, 1456, 1372, 1266, 1239, 1216, 1130, 1077, 1028 cm^{-1}; HR-MS: m/z = 272.1779, calcd. for $C_{18}H_{24}O_2$ [M^+]: 272.1776.

(R_a)-Benzyl 2-(4-chlorobutyl)-2,3-hexadienoate [(R_a)-2h]: The reaction of $Pd(dba)_2$ (17.3 mg, 0.03 mmol), (*S*)-SEGPHOS (18.6 mg, 0.03 mmol), $(NH_4)_2HPO_4$ (145.2 mg, 1.1 mmol), (*S*)-**1h** (253.0 mg, 1.0 mmol)/MTBE (1 mL), BnOH (1.0802 g, 10 mmol)/MTBE (1 mL) afforded (R_a)-**2h** (eluent: petroleum ether/ethyl ether = 30/1) as a liquid; yield: 256.8 mg (88%); 95% *ee*; HPLC [conditions: Regis (*S*,*S*) Whelk-O column, hexane/*i*-PrOH = 200/1, 0.5 mL min^{-1}, λ = 214 nm]: t_R (minor) = 28.4 min, t_R (major) = 32.0 min); $[\alpha]_D^{22}$: −29.3 (*c* 1.17, $CHCl_3$); ^1H NMR (300 MHz, $CDCl_3$): δ = 7.40–7.27 (m, 5H, Ar-H), 5.67–5.58 (m, 1H, =CH), 5.22 (d, *J* = 12.9 Hz, 1H, one proton from Bn), 5.15 (d, *J* = 12.9 Hz, 1H, one proton from Bn), 3.54 (t, *J* = 6.6 Hz, 2H, CH_2), 2.34–2.22 (m, 2H, CH_2), 2.20–2.07 (m, 2H, CH_2), 1.88–1.75 (m, 2H, CH_2), 1.67–1.53 (m, 2H, CH_2), 1.05 (t, *J* = 7.4 Hz, 3H, CH_3 from Et); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 209.7, 167.3, 136.3, 128.3, 127.8, 127.5, 100.5, 97.1, 66.1, 44.7, 31.9, 27.6, 25.3, 21.3, 13.2; MS (ESI): m/z = 317 [$M(^{37}Cl)+Na]^+$, 315 [$M(^{35}Cl)+Na]^+$; IR (neat): ν = 3066, 3034, 2963, 2934, 2871, 1955, 1707, 1498, 1455, 1375, 1242, 1191, 1144, 1095, 1048, 1011 cm^{-1}; HR-MS: m/z = 292.1232, calcd. for $C_{17}H_{21}O_2{}^{35}Cl$ [M^+]: 292.1230.

(R_a)-Benzyl 2-(2-phenylethyl)-2,3-hexadienoate [(R_a)-2i]: The reaction of $Pd(dba)_2$ (17.4 mg, 0.03 mmol), (*S*)-SEGPHOS (18.6 mg, 0.03 mmol), $(NH_4)_2HPO_4$ (145.4 mg, 1.1 mmol), (*S*)-**1i** (266.5 mg, 1.0 mmol)/MTBE (1 mL), BnOH (1.0803 g, 10 mmol)/MTBE (1 mL) afforded (R_a)-**2i** (eluent: petroleum ether/ethyl acetate = 30/1) as a liquid; yield: 256.3 mg (84%); 92% *ee*; HPLC [conditions: Regis (*S*,*S*) Whelk-O column, hexane/*i*-PrOH = 200/1, 0.5 mL min^{-1}, λ = 214 nm]: t_R (minor) = 29.2 min, t_R (major) = 32.3 min; $[\alpha]_D^{24}$: −13.8 (*c* 1.48, $CHCl_3$); ^1H NMR (300 MHz, $CDCl_3$): δ = 7.40–7.22 (m, 7H, Ar-H), 7.22–7.13 (m, 3H, Ar-H), 5.60–5.51 (m, 1H, =CH), 5.23 (d, *J* = 12.6 Hz, 1H, one proton from Bn), 5.16 (d, *J* = 12.9 Hz, 1H, one proton from Bn), 2.82–2.68 (m, 2H, CH_2), 2.67–2.47 (m, 2H, CH_2), 2.10–1.98 (m, 2H, CH_2), 0.97 (t, *J* = 7.4 Hz, 3H, CH_3 from Et); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 209.9, 167.2, 141.3, 136.3, 128.4, 128.3, 128.1, 127.8, 127.5, 125.8, 100.3, 97.0, 66.1, 34.3, 30.2, 21.2, 13.1; MS (ESI): m/z = 329 ($M+Na)^+$; IR (neat): ν = 3087, 3063, 3029, 2965, 2931, 2872, 1955, 1706, 1604, 1586, 1497, 1454, 1375, 1239, 1191, 1164, 1092, 1067, 1029 cm^{-1}; HR-MS: m/z = 306.1620, calcd. for $C_{21}H_{22}O_2$ [M^+]: 306.1620.

(R_a)-Benzyl 2-allyl-2,3-hexadienoate [(R_a)-2j]: The reaction of $Pd(dba)_2$ (17.3 mg, 0.03 mmol), (*S*)-SEGPHOS (18.4 mg, 0.03 mmol), $(NH_4)_2HPO_4$ (144.8 mg, 1.1 mmol), (*S*)-**1j** (201.4 mg, 1.0 mmol)/MTBE (1 mL), BnOH (1.0792 g, 10 mmol)/MTBE (1 mL) afforded (R_a)-**2j** (eluent: petroleum ether/ethyl ether = 30/1) as a liquid; yield: 205.9 mg (85%); 91% *ee*; HPLC [conditions: Regis (*S*,*S*) Whelk-O column, hexane/*i*-PrOH = 200/1, 0.5 mL min^{-1}, λ = 214 nm]: t_R (minor) = 19.2 min, t_R (major) = 22.0 min; $[\alpha]_D^{26}$: −38.0 (*c* 1.21, $CHCl_3$); ^1H NMR (300 MHz, $CDCl_3$): δ = 7.41–7.26 (m, 5H, Ar-H), 5.92–5.75 (m, 1H, CH=), 5.69–5.59 (m, 1H, =CH), 5.23 (d, *J* = 12.6 Hz, 1H, one proton from Bn), 5.16 (d, *J* = 12.9 Hz, 1H, one proton from Bn), 5.14–5.00 (m, 2H, =CH_2), 3.10–2.93 (m, 2H, CH_2), 2.20–2.06 (m, 2H, CH_2), 1.04 (t, *J* = 7.4 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ =

 © 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2013, 4, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201200910 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Advanced Synthesis & Catalysis

210.2, 167.1, 136.3, 135.2, 128.4, 127.9, 127.6, 116.0, 99.7, 97.2, 66.3, 33.1, 21.2, 13.1; MS (ESI): $m/z = 265$ $(M+Na)^+$, 243 $(M+H)^+$; IR (neat): $\nu = 3079$, 3068, 3035, 3007, 2968, 2935, 2911, 2876, 1957, 1709, 1455, 1376, 1270, 1246, 1222, 1122, 1065 cm^{-1}; HR-MS: $m/z = 224.1305$, calcd. for $C_{16}H_{18}O_2$ $[M^+]$: 242.1307.

(R_a)-Benzyl 2-(2-methylallyl)-2,3-hexadienoate [(R_a)-2k]: The reaction of $Pd(dba)_2$ (17.3 mg, 0.03 mmol), (*S*)-SEGPHOS (18.6 mg, 0.03 mmol), $(NH_4)_2HPO_4$ (145.0 mg, 1.1 mmol), (*S*)-**1k** (216.1 mg, 1.0 mmol)/MTBE (1 mL), BnOH (1.0800 g, 10 mmol)/MTBE (1 mL) afforded (R_a)-**2k** (eluent: petroleum ether/ethyl ether = 30/1) as a liquid; yield: 225.0 mg (88%); 95% *ee*; HPLC [conditions: Regis (*S*,*S*) Whelk-O column, hexane/*i*-PrOH = 200/1, 0.5 mL min^{-1}, $\lambda = 214$ nm]: t_R (minor) = 19.5 min, t_R (major) = 22.1 min; $[\alpha]_D^{22}$: −23.4 (*c* 2.30, $CHCl_3$); ^1H NMR (300 MHz, $CDCl_3$): $\delta = 7.41$–7.26 (m, 5H, Ar-H), 5.65–5.56 (m, 1H, =CH), 5.22 (d, $J = 12.9$ Hz, 1H, one proton from Bn), 5.15 (d, $J = 12.6$ Hz, 1H, one proton from Bn), 4.82–4.74 (m, 2H, CH_2 from $C=CH_2$), 3.04–2.90 (m, 2H, CH_2), 2.20–2.07 (m, 2H, CH_2), 1.74 (s, 3H, CH_3), 1.05 (t, $J = 7.2$ Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 210.5$, 167.3, 143.0, 136.4, 128.4, 127.9, 127.6, 111.8, 99.0, 96.6, 66.3, 37.4, 22.2, 21.3, 13.3; MS (ESI): $m/z = 279$ $(M+Na)^+$, 257 $(M+H)^+$; IR (neat): $\nu = 3073$, 3034, 2968, 2934, 2917, 2875, 1956, 1710, 1651, 1498, 1454, 1375, 1250, 1214, 1175, 1090, 1029 cm^{-1}; HR-MS: $m/z = 256.1464$, calcd. for $C_{17}H_{20}O_2$ $[M^+]$: 256.1463.

(R_a)-Benzyl 2-butyl-2,3-undecadienoate [(R_a)-2l]: The reaction of $Pd(dba)_2$ (17.3 mg, 0.03 mmol), (*S*)-SEGPHOS (18.5 mg, 0.03 mmol), $(NH_4)_2HPO_4$ (144.9 mg, 1.1 mmol), (*S*)-**1l** (287.9 mg, 1.0 mmol)/MTBE (1 mL), BnOH (1.0801 g, 10 mmol)/MTBE (1 mL) afforded (R_a)-**2l** (eluent: petroleum ether/ethyl ether = 30/1) as a liquid; yield: 279.4 mg (85%); 94% *ee*; HPLC [conditions: Regis (*S*,*S*) Whelk-O column, hexane/*i*-PrOH = 200/1, 0.5 mL min^{-1}, $\lambda = 214$ nm]: t_R (minor) = 17.2 min, t_R (major) = 20.0 min; $[\alpha]_D^{24}$: −42.6 (*c* 1.32, $CHCl_3$); ^1H NMR (300 MHz, $CDCl_3$): $\delta = 7.39$–7.27 (m, 5H, Ar-H), 5.57–5.48 (m, 1H, =CH), 5.21 (d, $J = 12.9$ Hz, 1H, one proton from Bn), 5.15 (d, $J = 12.9$ Hz, 1H, one proton from Bn), 2.30–2.15 (m, 2H, CH_2), 2.15–2.05 (m, 2H, CH_2), 1.50–1.16 (m, 14H, $7\times CH_2$), 0.96–0.83 (m, 6H, $2\times CH_3$); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 210.0$, 167.5, 136.4, 128.3, 127.7, 127.5, 100.4, 95.0, 66.0, 31.7, 30.2, 29.0, 28.9, 28.1, 28.0, 22.6, 22.2, 14.0, 13.8; MS (ESI): $m/z = 351$ $(M+Na)^+$, 329 $(M+H)^+$; IR (neat): $\nu = 3091$, 3066, 3034, 2956, 2926, 2857, 1956, 1711, 1456, 1263, 1236, 1217, 1119, 1071, 1029 cm^{-1}; HR-MS: $m/z = 328.2404$, calcd. for $C_{22}H_{32}O_2$ $[M^+]$: 328.2402.

(R_a)-Benzyl 2-butyl-6-methyl-2,3-heptadienoate [(R_a)-2m]: The reaction of $Pd(dba)_2$ (17.4 mg, 0.03 mmol), (*S*)-SEGPHOS (18.5 mg, 0.03 mmol), $(NH_4)_2HPO_4$ (144.8 mg, 1.1 mmol), (*S*)-**1m** (246.0 mg, 1.0 mmol)/MTBE (1 mL), BnOH (1.0795 g, 10 mmol)/MTBE (1 mL) afforded (R_a)-**2m** (eluent: petroleum ether/ethyl ether = 30/1) as a liquid; yield: 245.5 mg (86%); 97% *ee*; HPLC [conditions: Regis (*S*,*S*) Whelk-O column, hexane/*i*-PrOH = 400/1, 0.5 mL min^{-1}, $\lambda = 214$ nm]: t_R (minor) = 23.5 min, t_R (major) = 26.6 min; $[\alpha]_D^{24}$: −42.5 ($c = 1.13$, $CHCl_3$); ^1H NMR (300 MHz, $CDCl_3$) $\delta = 7.39$–7.27 (m, 5H, Ar-H), 5.52–5.42 (m, 1H, =CH), 5.20 (d, $J = 12.6$ Hz, 1H, one proton from Bn), 5.14 (d, $J = 12.6$ Hz, 1H, one proton from Bn), 2.29–2.18 (m, 2H, CH_2), 2.00 (t, $J = 7.1$ Hz, 2H, CH_2), 1.77–1.61 (m, 1H, CH from *i*-Bu), 1.49–1.25 (m, 4H, $2\times CH_2$), 0.96–0.85 (m, 9H, $3\times CH_3$); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta =$ 210.3, 167.5, 136.3, 128.3, 127.8, 99.7, 93.5, 66.2, 37.4, 30.2, 28.4, 28.1, 22.13, 22.07, 13.8; MS (ESI): $m/z = 309$ $(M+Na)^+$, 287 $(M+H)^+$; IR (neat): $\nu = 3091$, 3067, 3034, 2956, 2929, 2870, 1956, 1709, 1498, 1456, 1375, 1334, 1262, 1235, 1213, 1119, 1062, 1030, 1007 cm^{-1}; HR-MS: $m/z = 286.1936$, calcd. for $C_{19}H_{26}O_2$ $[M^+]$: 286.1933.

(R_a)-Benzyl 2-butyl-5-phenyl-2,3-pentadienoate [(R_a)-2n]: The reaction of $Pd(dba)_2$ (17.3 mg, 0.03 mmol), (*S*)-SEGPHOS (18.5 mg, 0.03 mmol), $(NH_4)_2HPO_4$ (145.4 mg, 1.1 mmol), (*S*)-**1n** (279.8 mg, 1.0 mmol)/MTBE (1 mL), BnOH (1.0804 g, 10 mmol)/MTBE (1 mL) afforded (R_a)-**2n** (eluent: petroleum ether/ethyl ether = 30/1) as a liquid; yield: 220.9 mg (69%); 96% *ee*; HPLC conditions: Regis (*S*,*S*) Whelk-O column, hexane/*i*-PrOH = 400/1, 0.5 mL min^{-1}, $\lambda = 214$ nm]: t_R (minor) = 45.5 min, t_R (major) = 48.6 min; $[\alpha]_D^{20}$: +2.7 (*c* 1.05, $CHCl_3$); ^1H NMR (300 MHz, $CDCl_3$): $\delta = 7.42$–7.30 (m, 5H, Ar-H), 7.23–7.16 (m, 5H, Ar-H), 5.72–5.62 (m, 1H, =CH), 5.24 (d, $J =$ 12.3 Hz, 1H, one proton from Bn), 5.15 (d, $J = 12.9$ Hz, 1H, one proton from Bn), 3.52–3.36 (m, 2H, CH_2), 2.31–2.15 (m, 2H, CH_2), 1.46–1.24 (m, 4H, $2\times CH_2$), 0.89 (t, $J = 7.1$ Hz, 3H, CH_3 from *n*-Bu); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 210.6$, 167.3, 139.1, 136.2, 128.54, 128.48, 128.3, 128.0, 127.9, 126.4, 100.9, 94.5, 66.4, 34.8, 30.2, 28.1, 22.2, 13.9; MS (ESI): $m/z =$ 343 $(M+Na)^+$; IR (neat): $\nu = 3087$, 3064, 3030, 2956, 2929, 2861, 1957, 1709, 1496, 1454, 1262, 1236, 1119, 1063 cm^{-1}; HR-MS: $m/z = 320.1775$, calcd. for $C_{22}H_{24}O_2$ $[M^+]$: 320.1776.

(R_a)-Benzyl 2-butyl-2,3,6-heptatrienoate [(R_a)-2o]: The reaction of $Pd(dba)_2$ (17.4 mg, 0.03 mmol), (*S*)-SEGPHOS (18.6 mg, 0.03 mmol), $(NH_4)_2HPO_4$ (144.8 mg, 1.1 mmol), (*S*)-**1o** (230.0 mg, 1.0 mmol)/MTBE (1 mL), BnOH (1.0807 g, 10 mmol)/MTBE (1 mL) afforded (R_a)-**2o** (eluent: petroleum ether/ethyl ether = 30/1) as a liquid; yield: 218.6 mg (81%); 96% *ee*; HPLC [conditions: Regis (*S*, *S*) Whelk-O column, hexane/*i*-PrOH = 200/1, 0.5 mL min^{-1}, $\lambda =$ 214 nm]: t_R (minor) = 18.9 min, t_R (major) = 21.8 min; $[\alpha]_D^{22}$: −41.6 (*c* 1.13, $CHCl_3$); ^1H NMR (300 MHz, $CDCl_3$): $\delta =$ 7.42–7.27 (m, 5H, Ar-H), 5.92–5.75 (m, 1H, =CH), 5.60–5.49 (m, 1H, =CH), 5.22 (d, $J = 12.6$ Hz, 1H, one proton from Bn), 5.19–5.07 (m, 2H, one proton from Bn and one proton from CH_2=), 5.07–5.00 (m, 1H, one proton from CH_2=), 2.86 (t, $J = 6.8$ Hz, 2H, CH_2), 2.34–2.16 (m, 2H, CH_2), 1.49–1.26 (m, 4H, $2\times CH_2$), 0.90 (t, $J = 7.1$ Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 210.4$, 167.4, 136.4, 135.3, 128.4, 127.9, 127.6, 116.1, 101.0, 93.1, 66.2, 32.3, 30.2, 28.1, 22.2, 13.9; MS (ESI): $m/z = 293$ $(M+Na)^+$, 271 $(M+H)^+$; IR (neat): $\nu = 3080$, 3067, 3034, 3006, 2957, 2929, 2872, 1957, 1711, 1456, 1376, 1263, 1237, 1120, 1063 cm^{-1}; HR-MS: $m/z = 270.1622$, calcd. for $C_{18}H_{22}O_2$ $[M^+]$: 270.1620.

(R_a)-Benzyl 2-*n*-heptyl-2,3-pentadienoate [(R_a)-2p]: The reaction of $Pd(dba)_2$ (17.3 mg, 0.03 mmol), (*S*)-SEGPHOS (18.5 mg, 0.03 mmol), $(NH_4)_2HPO_4$ (145.1 mg, 1.1 mmol), (*S*)-**1p** (247.0 mg, 1.0 mmol)/MTBE (1 mL), BnOH (1.0810 g, 10 mmol)/MTBE (1 mL) afforded (R_a)-**2p** (eluent: petroleum ether/ethyl ether = 40/1) as a liquid; yield: 244.1 mg (85%);: 93% *ee*; HPLC [conditions: Regis (*S*,*S*) Whelk-O column, hexane/*i*-PrOH = 200/1, 0.5 mL min^{-1}, $\lambda =$ 214 nm]: t_R (minor) = 19.2 min, t_R (major) = 22.8 min; $[\alpha]_D^{23}$: −27.0 ($c = 1.15$, $CHCl_3$) {80% *ee*, $[\alpha]_D$ −26.3 (*c* 0.32,

 © 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2013, 4, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201200910 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

$CHCl_3$)]};[17a] ^{1}H NMR (300 MHz, $CDCl_3$): δ 7.45–7.27 (m, 5H, Ar-H), 5.57–5.44 (m, 1H, =CH), 5.19 (s, 2H, CH_2 from Bn), 2.32–2.13 (m, 2H, CH_2), 1.76 (d, J=7.5 Hz, 3H, CH_3 from Me), 1.51–1.16 (m, 10H, 5×CH_2), 0.88 (t, J=6.3 Hz, 3H, CH_3 from n-C_7H_{15}); ^{13}C NMR (75 MHz, $CDCl_3$): δ= 210.7, 167.6, 136.5, 128.4, 127.8, 127.6, 100.0, 89.8, 66.2, 31.8, 29.1, 29.0, 28.4, 28.0, 22.6, 14.1, 13.3; MS (ESI): m/z=309 $(M+H)^+$; IR (neat): ν=2954, 2926, 2856, 1958, 1709, 1456, 1262, 1238, 1215, 1124, 1081 cm^{-1}.

(R_a)-Methyl 2-Butyl-2,3-hexadienoate and (*R*)-5-Ethyl-4-iodo-3-butylfuran-2(5*H*)-one

The reaction of $Pd(dba)_2$ (4.4 mg, 0.0075 mmol), (*S*)-SEGPHOS (4.8 mg, 0.0075 mmol), $(NH_4)_2HPO_4$ (35.9 mg, 0.275 mmol), (*S*)-**1a** (54.6 mg, 0.25 mmol)/MTBE (1 mL), MeOH (80.9 mg, 2.5 mmol)/MTBE (1 mL) afforded (R_a)-methyl 2-butyl-2,3-hexadienoate (eluent: petroleum ether/ethyl ether=20/1); yield: 32.3 mg (71%); $[\alpha]_D^{23}$: −38.7 (*c* 0.82, $CHCl_3$); ^{1}H NMR (300 MHz, $CDCl_3$): δ=5.61–5.52 (m, 1H, =CH), 3.71 (s, 3H, CH_3), 2.28–2.06 (m, 4H, 2×CH_2), 1.47–1.26 (m, 4H, 2×CH_2), 1.04 (t, J=7.4 Hz, 3H, CH_3), 0.89 (t, J=7.1 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ=209.5, 168.3, 101.0, 96.6, 52.0, 30.2, 28.2, 22.2, 21.4, 13.9, 13.2; MS (ESI *m/z*): 205 $(M+Na)^+$; IR (neat): ν=2959, 2931, 2873, 2862, 1957, 1714, 1436, 1378, 1267, 1243, 1190, 1121, 1077, 1017 cm^{-1}; HR-MS: m/z=182.1308, calcd. for $C_{11}H_{18}O_2$ $[M^+]$: 182.1307.

A solution of (R_a)-methyl 2-butyl-2,3-hexadienoate (20.9 mg, 0.12 mmol) and iodine (61.7 mg, 0.24 mmol) in MeCN/H_2O (15:1) (1 mL) was stirred at −15 °C for 11 h. The mixture was then quenched with a saturated aqueous solution of $Na_2S_2O_3$. This mixture was extracted with diethyl ether (10 mL×3), washed with brine (10 mL) and dried with anhydrous Na_2SO_4. Concentration and column chromatography on silica gel (petroleum ether/ethyl ether=40:1) afforded (*R*)-5-ethyl-4-iodo-3-butylfuran-2(5*H*)-one as a liquid; yield: 27.0 mg (80%); 80% *ee*; HPLC [conditions: Regis (*S*,*S*) Whelk-O column, hexane/*i*-PrOH=95/5, 0.5 mL min^{-1}, λ=254 nm]: t_R (minor)=16.8 min, t_R (major)=19.1 min); $[\alpha]_D^{23}$: +17.6 (*c* 0.57, $CHCl_3$); ^{1}H NMR (300 MHz, $CDCl_3$): δ=4.89–4.82 (m, 1H, CH), 2.38–2.29 (m, 2H, CH_2), 2.18–2.03 (m, 1H, one proton from CH_2), 1.75–1.62 (m, 1H, one proton from CH_2), 1.58–1.46 (m, 2H, CH_2), 1.42–1.28 (m, 2H, CH_2), 0.93 (t, J=7.5 Hz, 3H, CH_3), 0.90 (t, J=7.5 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ=170.2, 139.8, 120.7, 85.9, 29.2, 27.1, 25.6, 22.4, 13.8, 7.3; MS (EI, 70 eV): m/z (%)=294 (M^+, 2.14), 167 (100); IR (neat): ν=2958, 2929, 2873, 2860, 1750, 1636, 1458, 1329, 1267, 1201, 1115, 1048 cm^{-1}; HR-MS: m/z=294.0116, calcd. for $C_{10}H_{15}IO_2$ $[M^+]$: 294.0117.

(R_a)-Ethyl 2-Butyl-2,3-hexadienoate and (*R*)-5-Ethyl-4-iodo-3-butylfuran-2(5*H*)-one

The reaction of $Pd(dba)_2$ (4.3 mg, 0.0075 mmol), (*S*)-SEGPHOS (4.7 mg, 0.0075 mmol), $(NH_4)_2HPO_4$ (36.4 mg, 0.275 mmol), (*S*)-**1a** (53.9 mg, 0.25 mmol)/MTBE (1 mL), EtOH (114.6 mg, 2.5 mmol)/MTBE (1 mL) afforded (R_a)-ethyl 2-butyl-2,3-hexadienoate; yield: 35.4 mg (73%); $[\alpha]_D^{23}$: −39.8 (*c* 0.82, $CHCl_3$); ^{1}H NMR (300 MHz, $CDCl_3$): δ= 5.60–5.51 (m, 1H, =CH), 4.25–4.08 (m, 2H, CH_2), 2.27–2.04 (m, 4H, 2×CH_2), 1.47–1.19 (m, 7H, 2×CH_2 and CH_3), 1.04 (t, J=7.4 Hz, 3H, CH_3), 0.89 (t, J=7.1 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ=209.4, 167.8, 101.3, 96.5, 60.6, 30.3, 28.1, 22.2, 21.4, 14.2, 13.9, 13.2; MS (ESI): m/z= 219 $(M+Na)^+$; IR (neat): ν=2962, 2931, 2873, 1958, 1710, 1461, 1367, 1266, 1241, 1218, 1195, 1120, 1097, 1076, 1025 cm^{-1}; HR-MS: m/z=196.1462, calcd. for $C_{12}H_{20}O_2$ $[M^+]$: 196.1463.

A solution of (R_a)-ethyl 2-butyl-2,3-hexadienoate (27.9 mg, 0.14 mmol) and iodine (71.7 mg, 0.28 mmol) in MeCN/H_2O (15:1) (1 mL) was stirred at −15 °C for 11 h and afforded (*R*)-5-ethyl-4-iodo-3-butylfuran-2(5*H*)-one as a liquid; yield: 26.7 mg (64%); 84% *ee*; HPLC [conditions: Regis (*S*,*S*) Whelk-O column, hexane/*i*-PrOH=95/5, 0.5 mL min^{-1}, λ=254 nm]: t_R (minor)=16.8 min, t_R (major)=19.1 min; ^{1}H NMR (300 MHz, $CDCl_3$): δ=4.89–4.82 (m, 1H, CH), 2.38–2.29 (m, 2H, CH_2), 2.18–2.03 (m, 1H, one proton from CH_2), 1.75–1.62 (m, 1H, one proton from CH_2), 1.58–1.45 (m, 2H, CH_2), 1.42–1.28 (m, 2H, CH_2), 0.93 (t, J=7.4 Hz, 3H, CH_3), 0.90 (t, J=7.5 Hz, 3H, CH_3).

(R_a)-Isobutyl 2-Butyl-2,3-hexadienoate and (*R*)-5-Ethyl-4-iodo-3-butylfuran-2(5*H*)-one

The reaction of $Pd(dba)_2$ (4.2 mg, 0.0075 mmol), (*S*)-SEGPHOS (4.7 mg, 0.0075 mmol), $(NH_4)_2HPO_4$ (36.2 mg, 0.275 mmol), (*S*)-**1a** (54.6 mg, 0.25 mmol)/MTBE (1 mL), *i*-BuOH (185.2 mg, 2.5 mmol)/MTBE (1 mL) afforded (R_a)-isobutyl 2-butyl-2,3-hexadienoate (eluent: petroleum ether/ethyl ether=20/1) as a liquid; yield: 44.6 mg (79%); $[\alpha]_D^{23}$: −35.3 (*c* 1.05, $CHCl_3$); ^{1}H NMR (300 MHz, $CDCl_3$): δ= 5.58–5.49 (m, 1H, =CH), 3.97–3.88 (m, 1H, one proton from CH_2), 3.88–3.79 (m, 1H, one proton from CH_2), 2.32–2.16 (m, 2H, CH_2), 2.16–2.05 (m, 2H, CH_2), 2.00–1.84 (m, 1H, CH), 1.47–1.25 (m, 4H, 2×CH_2), 1.03 (t, J=7.4 Hz, 3H, CH_3), 0.94–0.84 (m, 9H, 3×CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ=209.6, 167.8, 101.3, 96.3, 70.6, 30.3, 28.1, 27.8, 22.2, 21.3, 19.0, 18.9, 13.8, 13.2; MS (ESI): m/z=471 $(2M+Na)^+$, 247 $(M+Na)^+$; IR (neat): ν=2961, 2932, 2874, 1957, 1710, 1466, 1376, 1265, 1240, 1121, 1076, 1015 cm^{-1}; HR-MS: m/z=224.1777, calcd. for $C_{14}H_{24}O_2$ $[M^+]$: 224.1776.

A solution of (R_a)-*iso*-butyl 2-butyl-2,3-hexadienoate (37.5 mg, 0.17 mmol) and iodine (87.4 mg, 0.34 mmol) in MeCN/H_2O (15:1) (1 mL) was stirred at −15 °C for 11 h and afforded (*R*)-5-ethyl-4-iodo-3-butylfuran-2(5*H*)-one as a liquid; yield: 43.5 mg (88%); 90% *ee*; HPLC [conditions: Regis (*S*,*S*) Whelk-O column, hexane/*i*-PrOH=95/5, 0.5 mL min^{-1}, λ=254 nm]: t_R (minor)=16.8 min, t_R (major)=19.2 min; ^{1}H NMR (300 MHz, $CDCl_3$): δ=4.89–4.82 (m, 1H, CH), 2.37–2.29 (m, 2H, CH_2), 2.18–2.03 (m, 1H, one proton from CH_2), 1.75–1.62 (m, 1H, one proton from CH_2), 1.57–1.45 (m, 2H, CH_2), 1.42–1.28 (m, 2H, CH_2), 0.93 (t, J=7.5 Hz, 3H, CH_3), 0.90 (t, J=7.5 Hz, 3H, CH_3).

(*R*)-5-Ethyl-4-iodo-3-methylfuran-2(5*H*)-one [(*R*)-3]

(1) A solution of (R_a)-**2b** (21.8 mg, 0.1 mmol) and iodine (51.4 mg, 0.2 mmol) in MeCN/H_2O (15:1) (1 mL) was stirred at −15 °C for 11 h. The mixture was then quenched with a saturated aqueous solution of $Na_2S_2O_3$. This mixture was extracted with diethyl ether (10 mL×3), washed with brine

 © 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2013, 4, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201200910 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

(10 mL) and dried with anhydrous Na_2SO_4. Concentration and column chromatography on silica gel (petroleum ether/ethyl ether = 40:1) afforded (*R*)-**3** as a white solid; yield: 21.3 mg (84%); mp 51–52 °C (petroleum ether/ethyl ether); >99% *ee* after recrystallization; HPLC [conditions: Regis (*S*,*S*) Whelk-O column, hexane/*i*-PrOH = 95/5, 0.5 mL min^{-1}, λ = 254 nm]: t_R (minor) = 22.5 min, t_R (major) = 26.5 min); $[\alpha]_D^{23}$: +31.1 (*c* 0.46, $CHCl_3$); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 4.90–4.82 (m, 1H, CH), 2.19–2.03 (m, 1H, one proton from CH_2), 1.94 (d, *J* = 1.8 Hz, 3H, CH_3), 1.74–1.58 (m, 1H, one proton from CH_2), 0.92 (t, *J* = 7.4 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 170.5, 136.1, 120.9, 86.2, 25.7, 12.8, 7.6; MS (EI 70 eV): *m/z* (%) = 252 (M^+, 19.65), 125 (100); IR (neat): ν = 2969, 2925, 2883, 2856, 1762, 1640, 1381, 1331, 1272, 1109, 1077, 1049, 1031 cm^{-1}; anal. calcd. for $C_7H_9IO_2$: C 33.36, H 3.60; found C 33.71, H 3.69.

(2) A solution of (R_a)-**2b** (10.4 mg, 0.05 mmol) and iodine (25.8 mg, 0.10 mmol) in MeCN/H_2O (15:1) (1 mL) was stirred at −15 °C for 11 h and afforded (*R*)-**3** as a white solid; yield: 10.7 mg (88%); 89% *ee*; HPLC [conditions: Regis (*S*,*S*) Whelk-O column, hexane/*i*-PrOH = 95/5, 0.5 mL min^{-1}, λ = 254 nm]: t_R (minor) = 21.5 min, t_R (major) = 24.6 min; ^{1}H NMR (300 MHz, $CDCl_3$): δ = 4.89–4.82 (m, 1H, CH), 2.18–2.03 (m, 1H, one proton from CH_2), 1.93 (d, *J* = 1.8 Hz, 3H, CH_3), 1.74–1.57 (m, 1H, one proton from CH_2), 0.92 (t, *J* = 7.4 Hz, 3H, CH_3).

Acknowledgements

Financial support from from National Natural Science Foundation of China (21232006) and State Basic Research Program of China (2011CB808700) is greatly appreciated. We also thank Zeng Rong in this group for reproducing the results of entries 4, 12 and 13 presented in Table 3.

References

[1] For recent reviews, see: a) A. S. K. Hashmi, *Angew. Chem.* **2000**, *112*, 3737; *Angew. Chem. Int. Ed.* **2000**, *39*, 3590; b) R. Zimmer, C. U. Dinesh, E. Nandanan, F. A. Khan, *Chem. Rev.* **2000**, *100*, 3067; c) S. Ma, *Acc. Chem. Res.* **2003**, *36*, 701; d) S. Ma, *Chem. Rev.* **2005**, *105*, 2829; e) S. Ma, *Aldrichimica Acta* **2007**, *40*, 91; f) S. Ma, *Acc. Chem. Res.* **2009**, *42*, 1679; g) B. Alcaide, P. Almendros, T. Martínez del Campo, *Chem. Eur. J.* **2010**, *16*, 5836; h) C. Aubert, L. Fensterbank, P. Garcia, M. Malacria, A. Simonneau, *Chem. Rev.* **2011**, *111*, 1954; i) F. Inagaki, S. Kitagaki, C. Mukai, *Synlett* **2011**, 594; j) F. López, J. L. Mascareñas, *Chem. Eur. J.* **2011**, *17*, 418.

[2] a) Y. Li, H. Zou, J. Gong, J. Xiang, T. Luo, J. Quan, G. Wang, Z. Yang, *Org. Lett.* **2007**, *9*, 4057; b) G. Chai, S. Wu, C. Fu, S. Ma, *J. Am. Chem. Soc.* **2011**, *133*, 3740; c) T. J. Martin, V. G. Vakhshori, Y. S. Tran, O. Kwon, *Org. Lett.* **2011**, *13*, 2586.

[3] P. de March, J. Font, A. Gracia, Q. Zheng, *J. Org. Chem.* **1995**, *60*, 1814.

[4] a) B. S. Santos, A. L. Cardoso, A. M. Beja, M. R. Silva, J. A. Paixão, F. Palacios, T. M. V. D. Pinho e Melo, *Eur. J. Org. Chem.* **2010**, 3249; b) B. J. Cowen, L. B. Saunders, S. J. Miller, *J. Am. Chem. Soc.* **2009**, *131*, 6105; c) X. Guan, Y. Wei, M. Shi, *J. Org. Chem.* **2009**, *74*, 6343.

[5] T. H. Lambert, D. W. C. MacMillan, *J. Am. Chem. Soc.* **2002**, *124*, 13646.

[6] a) J. Li, Y. Liu, C. Li, X. Jia, *Adv. Synth. Catal.* **2011**, *353*, 913; b) S. Xu, L. Zhou, R. Ma, H. Song, Z. He, *Chem. Eur. J.* **2009**, *15*, 8698; c) L. Rout, A. M. Harned, *Chem. Eur. J.* **2009**, *15*, 12926; d) Q. Zhang, L. Yang, X. Tong, *J. Am. Chem. Soc.* **2010**, *132*, 2550; e) Y. Fujiwara, G. C. Fu, *J. Am. Chem. Soc.* **2011**, *133*, 12293; f) H. Guo, Q. Xu, O. Kwon, *J. Am. Chem. Soc.* **2009**, *131*, 6318; g) H. Clavier, K. L. Jeune, I. de Riggi, A. Tenaglia, G. Buono, *Org. Lett.* **2011**, *13*, 308.

[7] a) B. Chen, S. Ma, *Chem. Eur. J.* **2011**, *17*, 754; b) M. N. Hopkinson, A. Tessier, A. Salisbury, G. T. Giuffredi, L. E. Combettes, A. D. Gee, V. Gouverneur, *Chem. Eur. J.* **2010**, *16*, 4739; c) M. N. Hopkinson, J. E. Ross, G. T. Giuffredi, A. D. Gee, V. Gouverneur, *Org. Lett.* **2010**, *12*, 4904; d) Y. Shi, K. E. Roth, S. D. Ramgren, S. A. Blum, *J. Am. Chem. Soc.* **2009**, *131*, 18022; e) A. S. K. Hashmi, L. Christian, D. René, R. Matthias, D. R. Tanuja, R. Frank, *Angew. Chem.* **2009**, *121*, 8392; *Angew. Chem. Int. Ed.* **2009**, *48*, 8243.

[8] Y. Deng, X. Jin, C. Fu, S. Ma, *Org. Lett.* **2009**, *11*, 2169.

[9] For recent reviews see: a) S. Yu, S. Ma, *Chem. Commun.* **2011**, *47*, 5384; b) K. M. Brummond, J. E. Deforrest, *Synthesis* **2007**, 795.

[10] M. Ogasawara, *Tetrahedron: Asymmetry* **2009**, *20*, 259.

[11] For isomerizations of alkynoates affording chiral 2,3-allenoates, see: H. Liu, D. Leow, K.-W. Huang, C.-H. Tan, *J. Am. Chem. Soc.* **2009**, *131*, 7212.

[12] For kinetic resolution of racemic 2,3-allenoates affording chiral 2,3-allenoates, see: J. Yu, W.-J. Chen, L.-Z. Gong, *Org. Lett.* **2010**, *12*, 4050.

[13] For asymmetric Wittig reaction affording chiral 2,3-allenoates, see: a) T. M. V. D. Pinho e Melo, A. L. Cardoso, A. M. d′A. Rocha Gonsalves, J. C. Pessoa, J. A. Paixão, A. M. Beja, *Eur. J. Org. Chem.* **2004**, 4830; b) C.-Y. Li, X.-B. Wang, X.-L. Sun, Y. Tang, J.-C. Zheng, Z.-H. Xu, Y.-G. Zhou, L.-X. Dai, *J. Am. Chem. Soc.* **2007**, *129*, 1494; c) J. Yamazaki, T. Watanabe, K. Tanaka, *Tetrahedron: Asymmetry* **2001**, *12*, 669.

[14] For asymmetric selenoxide elimination affording chiral 2,3-allenoates, see: Y. Nishibayashi, J. D. Singh, S. Fukuzawa, S. Uemura, *J. Org. Chem.* **1995**, *60*, 4114.

[15] For dynamic kinetic protonation of racemic allenylmetal species affording chiral 2,3-allenoates, see: K. Mikami, A. Yoshida, *Angew Chem.* **1997**, *109*, 892; *Angew. Chem. Int. Ed. Engl.* **1997**, *36*, 858.

[16] For cross-coupling reaction of diethyl (*S*)-1-(methoxycarbonyl)-1,6-diphenylhex-2-yn-1-yl phosphate affording chiral 2,3-allenoates, see: G. A. Molander, E. M. Sommers, S. R. Baker, *J. Org. Chem.* **2006**, *71*, 1563.

[17] For alkoxycarbonylation of allenyl Pd intermediates affording chiral 2,3-allenoates, see: a) J. A. Marshall, M. A. Wolf, *J. Org. Chem.* **1996**, *61*, 3238; b) J. A. Marshall, G. S. Bartley, E. M. Wallace, *J. Org. Chem.* **1996**, *61*, 5729; c) J. A. Marshall, M. A. Wolf, E. M. Wallace, *J. Org. Chem.* **1997**, *62*, 367.

 © 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2013, 4, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201200910 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

[18] For S_N2' reaction affording chiral 2,3-allenoates, see: K. Kobayashi, H. Naka, A. E. H. Wheatley, Y. Kondo, *Org. Lett.* **2008**, *10*, 3375.
[19] C. Fu, S. Ma, *Eur. J. Org. Chem.* **2005**, 3942.
[20] The crystal data of compound (*R*)-**3:** $C_7H_9IO_2$; MW = 252.04, triclinic, monoclinic, space group *P*2(1)/*n*, Mo Kα, final *R* indices [$I > 2\sigma(I)$], *R*1 = 0.0204, *wR*2 = 0.0490, *a* = 8.134(3) Å, *b* = 8.334(3) Å, *c* = 12.917(4) Å, $\alpha = 90°$, $\beta = 90°$, $\gamma = 90°$, *V* = 875.6(5) Å3, *T* = 293(2) K, *Z* = 4, reflections collected/unique: 4570/1714 (R_{int} = 0.0234), number of observations [$> 2\sigma(I)$] 1689, parameters 94. CCDC 887542 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre *via* www.ccdc.cam.ac.uk/data_request/cif.
[21] a) M. Hidai, M. Kokura, Y. Uchida, *J. Organomet. Chem.* **1973**, *52*, 431; b) A. Schoenberg, I. Bartoletti, R. F. Heck, *J. Org. Chem.* **1974**, *39*, 3318; c) N. Miyaura, A. Suzuki, *Chem. Lett.* **1981**, 879; d) F. Ozawa, N. Kawasaki, H. Okamoto, T. Yamamoto, A. Yamamoto, *Organometallics* **1987**, *6*, 1640; e) D. Milstein, *Acc. Chem. Res.* **1988**, *21*, 428.

 © 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2013, 4, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201200910 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License